

ENVS193DS-Final

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Repository Link

[GitHub Repository](#)

Set Up

Packages:

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(here)
library(scales)
library(ggeffects)
library(MuMIn)
library(forcats)
library(DHARMA)
library(ggeffects)
library(gtsummary)
```

Data:

```
# creating object from 'nests' from data
nests <- read.csv(here('data', 'nest_data_final.csv'))

# creating object 'mz_data' from data
mz_data <- read_xlsx(here('data', '193DS-personal-data-updated.xlsx'))
```

Problem 1

a.

In part 1 of the results section, the co-worker used a linear regression. In part 2 of the results section, the co-worker used a Chi-Squared test.

b.

Statement 1: Based collected American Avocet micro-habitat use and distance to water edge, American Avocet micro-habitat use is significantly influenced by proximity to water edge.

Statement 2: Based on collected shorebird species presence across the habitats managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflap, shorebird species significantly differ by locale.

c.

One additional variable that could influence the probability of American Avocet microhabitat use is the water depth surrounding habitats, a continuous variable measured in meters, collected via bathymetry or hands-on-measurement depending on depth. This variable might influence American Avocet habitat use as water depth is a significant driver of bird use in managed wetlands – water depth may influence American Avocet use depending on feeding patterns (deeper areas are inaccessible to wading birds), present food type (different depths may lead to different flora and fauna presence), and predator presence (predators may reside in varying depths of water/predators may not be able to access certain water depths), playing a role in determining how American Avocets use these locales (Schacter et. al., 11).

Problem 2

a.

```
# create new object called 'nests_clean' from 'nests'
nests_clean <- nests |>
  # clean names
  clean_names() |>
  # filter ant species
  filter(
    ant_species %in% c("RRB", "AB", "BBR")) |>
  # full names
  mutate(
    species_name_full = recode_values(ant_species,
                                      "RRB" ~ "Crematogaster mimosae",
                                      "AB" ~ "Crematogaster sjostedti",
                                      "BBR" ~ "Crematogaster nigriceps")) |>

# reordering species
mutate(
```

```

# order of species
species_name_full = as_factor(species_name_full),
species_name_full = fct_relevel(species_name_full,
                                "Crematogaster mimosae",
                                "Crematogaster sjostedti",
                                "Crematogaster nigriceps")) |>

# arrange in order
arrange(species_name_full) |>
# columns of interest
select(
  height_cm,
  case_control,
  species_name_full)

# display structure
str(nests_clean)

```

```

'data.frame':  270 obs. of  3 variables:
 $ height_cm      : num  392 310 513 154 413 ...
 $ case_control   : int   1 0 0 0 1 0 0 0 0 0 ...
 $ species_name_full: Factor w/ 3 levels "Crematogaster mimosae",...: 1 1 1 1 1 1 1 1 1 1 ...

```

```

# display 10 random rows
nests_clean |> slice_sample(n = 10)

```

	height_cm	case_control	species_name_full
1	150.0	0	Crematogaster mimosae
2	391.5	1	Crematogaster mimosae
3	228.0	0	Crematogaster mimosae
4	406.0	0	Crematogaster mimosae
5	130.0	0	Crematogaster mimosae
6	359.5	0	Crematogaster mimosae
7	181.0	1	Crematogaster mimosae
8	104.0	0	Crematogaster mimosae
9	427.0	0	Crematogaster mimosae
10	166.5	0	Crematogaster mimosae

b.

In this data set, 1s and 0s indicate if a tree has a nest (1) or is “available” (0); in other words, they indicate if a tree contains a birds nest or not.

c.

Tree Height: Tree height is a continuous variable, measured in centimeters. According to paragraph three in the introduction subsection, as well as paragraph two in the results subsection of “Symbiotic acacia ants drive nesting behavior by birds in an African savanna”, as tree height increases, probability of nest occurrence probability will increase as well, as higher nests decrease ground predator nest predation rates, increasing overall bird lifetime fitness.

Acacia Ant Species: Acacia ant species is a categorical variable (C.nigriceps, C.mimosae, and C.sjostedti). According to paragraph three in the introduction subsection in “Symbiotic acacia ants drive nesting behavior by birds in an African savanna”, the presence of more aggressive defending symbionts (C.nigriceps and C.mimosae) will result in an increased probability of nest occurrence probability, aggressive defender presence may protect nests, reducing nest predation rates, and increase overall bird lifetime fitness.

d.

Model number	Ant Species	Tree Height	Predictor List
0 (null)			no predictors
1 (full model)	x	x	all predictors
2 (species)	x		species
3 (height)		x	tree height
4 (interact)	x	x	all predictors – interacting

e.

```
# model 0: null model
model_0 <- glm(
  # formula
  case_control ~ 1,
  # data frame
  data = nests_clean,
  family = "binomial")

# model 1: no interaction
model_1 <- glm(
  # formula
  case_control ~ height_cm + species_name_full,
  # data frame
```

```

    data = nests_clean,
    family = "binomial")

# model 2: ant species
model_2 <- glm(
  case_control ~ species_name_full,
  # data frame
  data = nests_clean,
  family = "binomial")

# model 3: tree height
model_3 <- glm(
  # formula
  case_control ~ height_cm,
  # data frame
  data = nests_clean,
  family = "binomial")

# model 4: interaction
model_4 <- glm(
  # formula
  case_control ~ height_cm * species_name_full,
  # data frame
  data = nests_clean,
  family = "binomial")

```

f.

```

# running AICc
AICc(
  model_0,
  model_1,
  model_2,
  model_3,
  model_4
) |>
# display ordered results
arrange(AICc)

```

df	AICc
----	------

```

model_4 6 238.1236
model_1 4 258.8940
model_3 2 266.1994
model_2 3 287.1103
model_0 1 288.0563

```

The best model as determined by Akaike's Information Criterion (AIC) was the model using interacting predictor variables ant species and tree height for the response variable of bird nest presence (model 4).

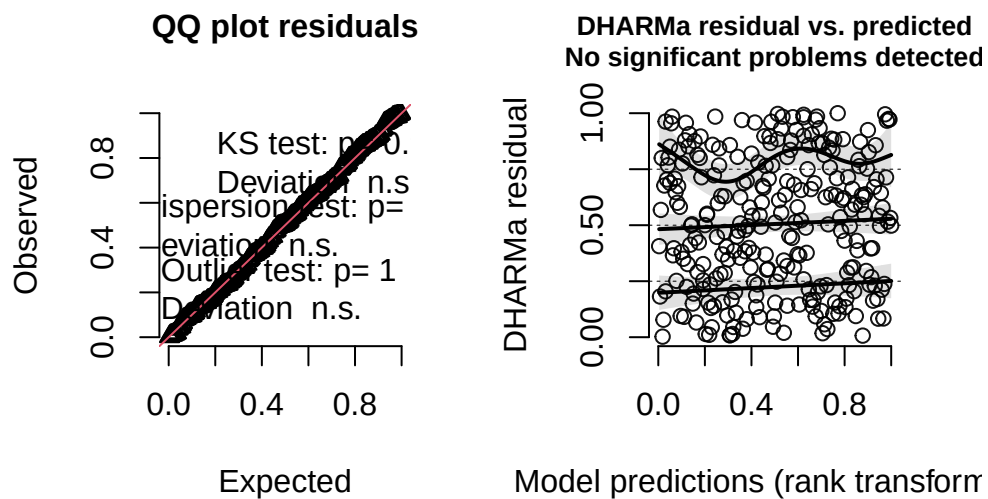
g.

```

# run dharma residuals
plot(simulateResiduals(model_4))

```

DHARMa residual



h.

```

# generating all predictions
predictions <- ggpredict(
  model_4,
  terms = c("height_cm [all]",
            "species_name_full")) |>
# renaming columns
rename(height_cm = x,
       species_name_full = group)

```

```

# base layer: ggplot
# use nests_clean data
ggplot(data = nests_clean,
      # x-axis
      aes(x = height_cm,
          # y-axis
          y = case_control)) +
# first layer: points
# color by species
geom_point(aes(color = species_name_full),
          # change point shape
          shape = 21) +
# second layer: line
# use predictions data
geom_line(data = predictions,
          # x-axis
          aes(x = height_cm,
              # y-axis
              y = predicted,
              # color by species
              color = species_name_full),
          # set linewidth
          linewidth = 2) +
# third layer: ribbon
# use predictions data
geom_ribbon(data = predictions,
          # x-axis
          aes(x = height_cm,
              # y-axis
              y = predicted,
              # y-min
              ymin = conf.low,
              # y-max

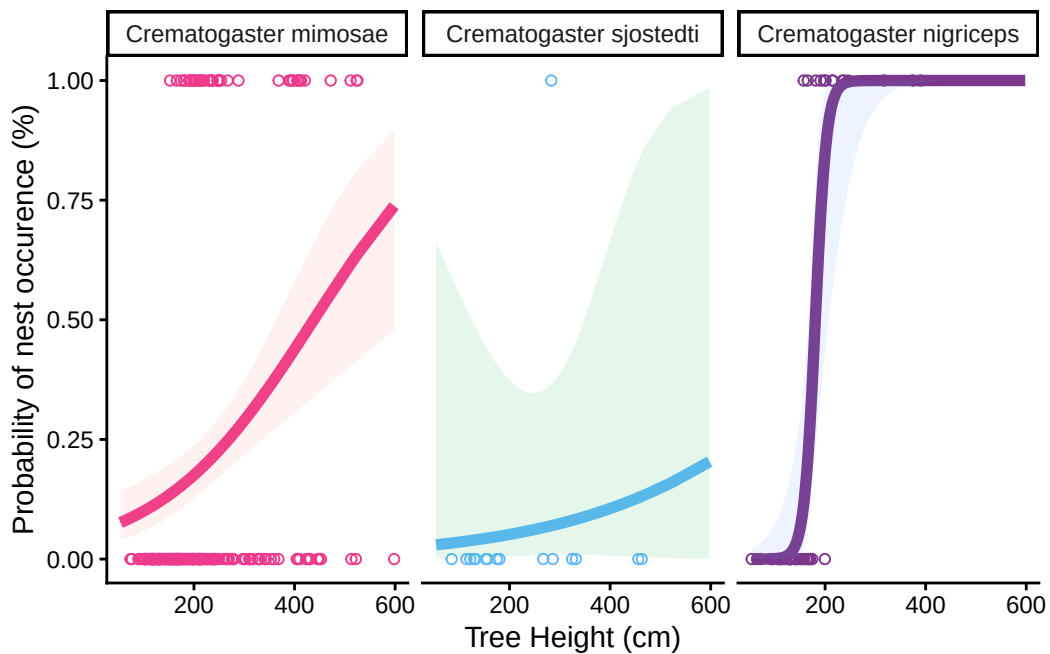
```



```

      ymax = conf.high,
      # fill by species
      fill = species_name_full),
      # change opacity
      alpha = 0.1) +
# panels by species
facet_wrap(vars(species_name_full)) +
labs(
  # label x-axis
  x = "Tree Height (cm)",
  # label y-axis
  y = "Probability of nest occurrence (%)") +
# color by species
scale_color_manual(
  values = c("Crematogaster mimosae" = "violetred2",
             "Crematogaster sjostedti" = "steelblue1",
             "Crematogaster nigriceps" = "mediumorchid4")) +
# use classic theme
theme_classic() +
# remove legend
theme(legend.position = "none")

```



i.

Figure 1. Aggressive ant species tree presence and increasing tree height increase nest presence probability. Points represent observations of individual nest occurrences (0||1) with their respective tree heights (cm). Solid lines represent model predictions for nest presence, with shaded ribbon areas indicating the 95% confidence interval range. Colors represent ant species (pink: *C. mimosae*, blue: *C. sjostedti*, purple: *C. nigriceps*). Data originally from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: “Symbiotic acacia ants drive nesting behavior by birds in an African savanna” [Data set]. *Zenodo*. <https://doi.org/10.5281/zenodo.8373322>

j.

```
# generating prediction
ggpredict(model_4,
  # setting terms
  terms = c("height_cm [600]",
            "species_name_full"))
```

```
# Predicted probabilities of case_control
```

```
species_name_full: Crematogaster mimosae
```

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

```
species_name_full: Crematogaster sjostedti
```

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

```
species_name_full: Crematogaster nigriceps
```

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

k.

Ant species presence and tree height interact to predict probability of bird presence. As ant aggressiveness and tree height increase, the probability of nest presence increases. At 600 cm, nest presence probability is 0.74 (95% CI:[0.48, 0.90]) for *C. mimosae*, 0.20 (95% CI:[0.00, 0.99]) for *C. sjostedti*, and 1.00 (95% CI:[1.00, 1.00]) for *C. nigriceps*; bird nest occurrence occurs at higher rates in trees with more aggressive acacia ant species, indicating that more aggressive symbionts may confer additional protections from predators beyond residing trees to bird nests, reducing nest predation rates and increasing overall bird lifetime fitness.

Problem 3

a.

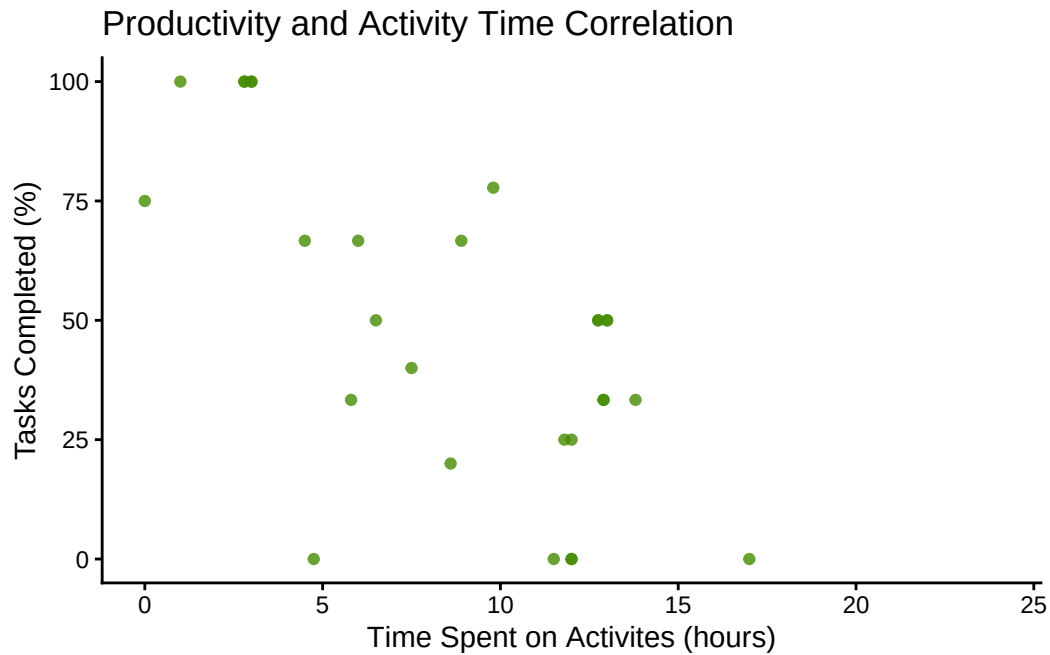
My visualizations used different metrics and different model types across Homework 2/3 and my affective visualization; in Homework 2 and 3 I used box plots to visualize my data, while I used a trend line to visualize my data in my affective visualization. Across my visualizations, I did not see many similarities – my affective visualization reflected my predicted outcomes using variables that I did not use in my 2/3 visualizations; subsequently, I did not see any patterns across my affective and effective visualizations, likely due to different metrics of creation and data alignment. I did not receive any feedback on my Homework 2 and 3 visualizations, however, in workshop, I got a lot of visual suggestions – for instance, implementing a clearer trend line in my drawing – which was reflected in my final affective visualization (I added a red trend line, and used different color schemes to create clearer visual distinctions between my data).

b.

Using the response variable of productivity % and the predictor variable of activity time (as expressed in my affective visualization as an aggregation of outside activity and class time), one analysis that we talked about this quarter that I could apply to my research question is a correlation analysis, specifically a Spearman's correlation. Both of my variables are independent numerical continuous variables, with one variable (activity time) predicted to have an influence on the other variable (productivity %), suspected monotonic relationship. Additionally, there are not that many observations in data. Subsequently, I believe that a Spearman's correlation analysis is appropriate in trying to answer the question: how do my activities impact my productivity?

C.

```
# base layer: ggplot
ggplot(
  # starting data frame
  data = mz_clean,
  # x-axis
  mapping = aes(x = total_activity_time,
                 # y-axis
                 y = completed_v_intended)
) +
  # first layer: points
  geom_point(
    # coloring points
    color = "chartreuse4",
    # changing opacity
    alpha = 0.8
  ) +
  # relabelling x-axis, y-axis, and title
  labs(
    # label x-axis
    x = "Time Spent on Activites (hours)",
    # label y-axis
    y = "Tasks Completed (%)",
    # add chart title
    title = 'Productivity and Activity Time Correlation') +
  # use classic theme
  theme_classic() +
  # remove legend
  theme(legend.position = "none")
```



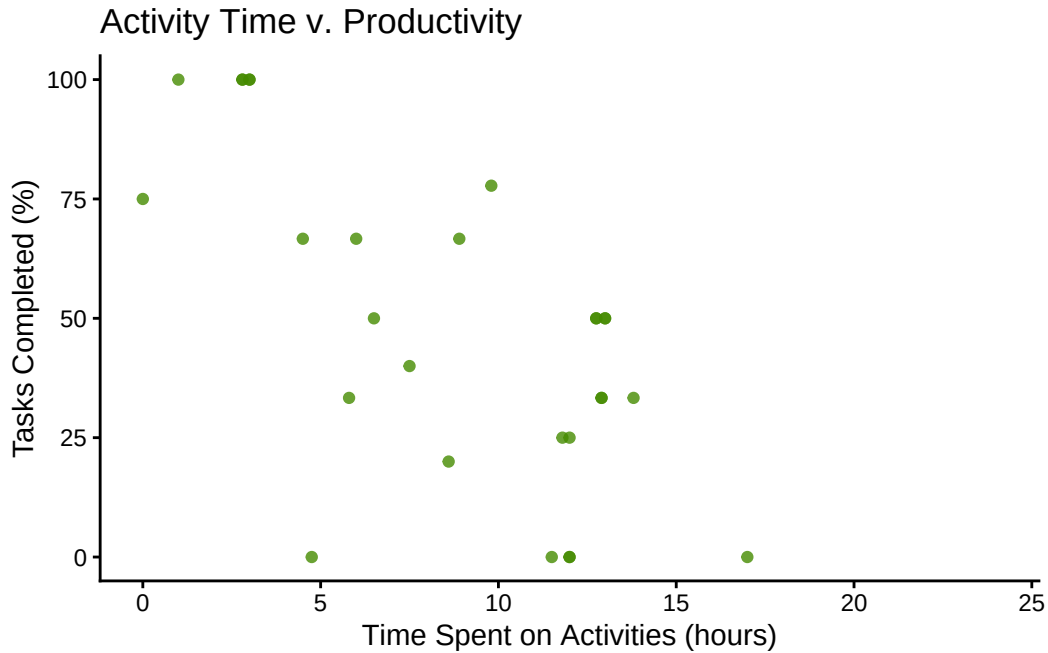
```
cor.test(
  # y-variable
  mz_clean$completed_v_intended,
  # x-variable
  mz_clean$total_activity_time,
  # use spearman's test
  method = 'spearman',
  # use normal p-approximation
  exact = FALSE)
```

Spearman's rank correlation rho

```
data:  mz_clean$completed_v_intended and mz_clean$total_activity_time
S = 5787.6, p-value = 0.001106
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.5839114
```

d.

```
# base layer: ggplot
ggplot(
  # starting data frame
  data = mz_clean,
  # x-axis
  mapping = aes(x = total_activity_time,
                 # y-axis
                 y = completed_v_intended)
) +
  # first layer: points
  geom_point(
    # coloring points
    color = "chartreuse4",
    # changing opacity
    alpha = 0.8
  ) +
  # relabelling x-axis, y-axis, and title
  labs(
    # label x-axis
    x = "Time Spent on Activities (hours)",
    # label y-axis
    y = "Tasks Completed (%)",
    # add chart title
    title = 'Activity Time v. Productivity') +
  # use classic theme
  theme_classic() +
  # remove legend
  theme(legend.position = "none")
```



e.

Figure 2. Activity time decreases productivity. Small descending round solid green points represent individual measurements of productivity (% of tasks completed) in relation to time spent that day on activities (hours). Data from 193DS-personal-data-updated (Zhao M, 2026).

f.

To evaluate the strength of the relationship between activity time and productivity I used a Spearman's rank correlation test. I chose to use Spearman's test over Pearson's as I did not check for normality nor a linear relationship. After running the Spearman's test, I found a moderate negative relationship between ocean temperature and giant kelp frond elongation rate (Spearman $\rho = -0.58$, $S = 5787.6$, $p = 0.001$, $\alpha = 0.05$). As time spent on activities increases, productivity decreases.

g.

Completed in section